

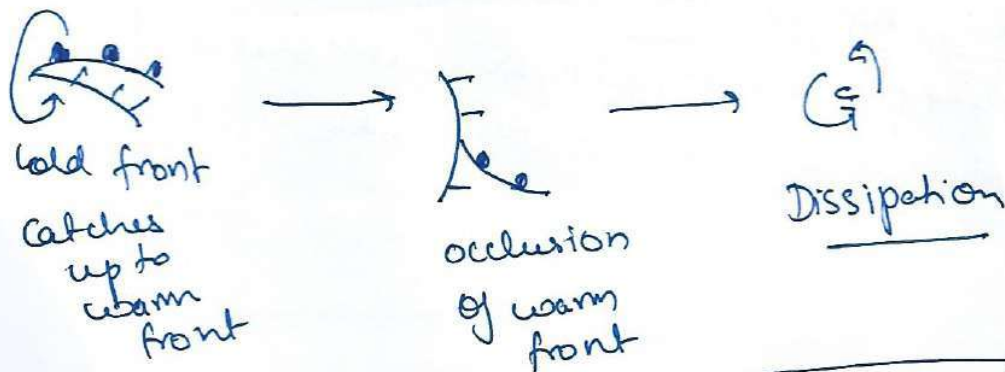
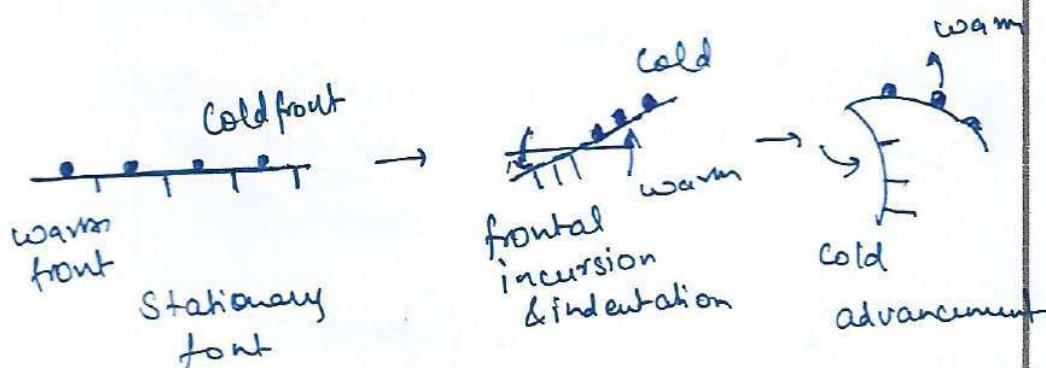


Q.16) Explain and illustrate the life-cycle of a temperate cyclone. Compare the structure and associated weather conditions of tropical cyclone with that of temperate cyclone. (15 Marks, 250 Words)

समशीतोष्ण चक्रवात के जीवन-चक्र की व्याख्या कीजिए। समशीतोष्ण चक्रवात के साथ उष्णकटिबंधीय चक्रवात की संरचना और संबद्ध मौसम की स्थिति की तुलना करें। (15 अंक, 250 शब्द)

Temperate cyclones are extra-tropical cyclones created in the temperate region ($30-35^{\circ}$ N and S) due to frontogenesis.

Life cycle of temperate cyclone



III Cold
● warm

The temperate cyclone is thus formed when two different fronts meet and collide with each other.

It is sustained due to the difference in airmass temperatures, and densities of the two fronts.

It finally dissipates ~~one~~ once the cold front captures the warm front as shown in the figure.

Tropical Temperate Cyclone	Temperate Tropical Cyclone
Location: Tropics	Location: Temperate regions.
Structure: Narrower width (100-500 km)	Structure: wider expanse. (300-2000 km)
Height: Taller than temperate cyclones	It is of shorter vertical expanse.
Lasts for <u>1-2 weeks</u>	Lasts for <u>longer durations</u> (2-3 weeks)

- Causes sudden intense rainfall
- Causes more destruction due to high speed winds
- Structurally moves from east to west (trade winds)
- Consists of a calm eye region
- Clouds involved - nimbostratus

Influence on India

- Both coasts affected

- Causes lighter rainfall for longer duration
- Destruction is more due to the ensuing floods

Moves from west to east (westerlies)

- ~~Go~~ Does not have a calm region
- Clouds involved - cumulonimbus and of more variety
- Western Disturbances in north-west India

Cyclones are thus a potent disaster which needs to be catered to through effective implementation of NDMA guidelines.

Feedback (For OFFICE use only)

Structure		Content	
Question Interpretation		Total :	